

```
In [27]: import numpy as np
import pandas as pd
from scipy.stats import mode
import matplotlib.pyplot as plt
import seaborn as sns
```

QUESTION 1

```
In [3]: df = pd.read_csv('food.csv')
```

```
In [4]: df
```

```
Out[4]:
```

	Name	Ingredients	Diet	cook_time	flavor_profile	course	state	region
0	Balu shahi	Maida flour, yogurt, oil, sugar	vegetarian	25	sweet	dessert	West Bengal	East
1	Boondi	Gram flour, ghee, sugar	vegetarian	30	sweet	dessert	Rajasthan	West
2	Gajar ka halwa	Carrots, milk, sugar, ghee, cashews, raisins	vegetarian	60	sweet	dessert	Punjab	North
3	Ghevar	Flour, ghee, kewra, milk, clarified butter, su...	vegetarian	30	sweet	dessert	Rajasthan	West
4	Gulab jamun	Milk powder, plain flour, baking powder, ghee,...	vegetarian	40	sweet	dessert	West Bengal	East
...	...	...	...	...	...	...	...	...
248	Til Pitha	Glutinous rice, black sesame seeds, gur	vegetarian	30	sweet	dessert	Assam	North East
249	Bebinca	Coconut milk, egg yolks, clarified butter, all...	vegetarian	60	sweet	dessert	Goa	West
250	Shufta	Cottage cheese, dry dates, dried rose petals, ...	vegetarian	-1	sweet	dessert	Jammu & Kashmir	North
251	Mawa Bati	Milk powder, dry fruits, arrowroot powder, all...	vegetarian	45	sweet	dessert	Madhya Pradesh	Central
252	Pinaca	Brown rice, fennel seeds, grated coconut, blac...	vegetarian	-1	sweet	dessert	Goa	West

253 rows × 8 columns

```
In [5]: df.columns
```

```
Out[5]: Index(['Name', 'Ingredients', 'Diet', 'cook_time', 'flavor_profile', 'course',  
          'state', 'region'],  
          dtype='object')
```

```
In [8]: df.describe()
```

```
Out[8]:
```

	cook_time
count	253.000000
mean	37.604743
std	47.515765
min	-1.000000
25%	20.000000
50%	30.000000
75%	45.000000
max	720.000000

```
In [9]: df.head()
```

```
Out[9]:
```

	Name	Ingredients	Diet	cook_time	flavor_profile	course	state	region
0	Balu shahi	Maida flour, yogurt, oil, sugar	vegetarian	25	sweet	dessert	West Bengal	East
1	Boondi	Gram flour, ghee, sugar	vegetarian	30	sweet	dessert	Rajasthan	West
2	Gajar ka halwa	Carrots, milk, sugar, ghee, cashews, raisins	vegetarian	60	sweet	dessert	Punjab	North
3	Ghevar	Flour, ghee, kewra, milk, clarified butter, su...	vegetarian	30	sweet	dessert	Rajasthan	West
4	Gulab jamun	Milk powder, plain flour, baking powder, ghee,...	vegetarian	40	sweet	dessert	West Bengal	East

```
In [10]: df.tail()
```

Out[10]:

	Name	Ingredients	Diet	cook_time	flavor_profile	course	state	region
248	Til Pitha	Glutinous rice, black sesame seeds, gur	vegetarian	30	sweet	dessert	Assam	North East
249	Bebinca	Coconut milk, egg yolks, clarified butter, all...	vegetarian	60	sweet	dessert	Goa	West
250	Shufta	Cottage cheese, dry dates, dried rose petals, ...	vegetarian	-1	sweet	dessert	Jammu & Kashmir	North
251	Mawa Bati	Milk powder, dry fruits, arrowroot powder, all...	vegetarian	45	sweet	dessert	Madhya Pradesh	Central
252	Pinaca	Brown rice, fennel seeds, grated coconut, blac...	vegetarian	-1	sweet	dessert	Goa	West

In [12]:

```
df['state'] = df['state'].replace('-1', pd.NA).mode()[0]
```

In [13]:

```
df
```

Out[13]:

	Name	Ingredients	Diet	cook_time	flavor_profile	course	state	region
0	Balu shahi	Maida flour, yogurt, oil, sugar	vegetarian	25	sweet	dessert	Gujarat	East
1	Boondi	Gram flour, ghee, sugar	vegetarian	30	sweet	dessert	Gujarat	West
2	Gajar ka halwa	Carrots, milk, sugar, ghee, cashews, raisins	vegetarian	60	sweet	dessert	Gujarat	North
3	Ghevar	Flour, ghee, kewra, milk, clarified butter, su...	vegetarian	30	sweet	dessert	Gujarat	West
4	Gulab jamun	Milk powder, plain flour, baking powder, ghee,...	vegetarian	40	sweet	dessert	Gujarat	East
...	...	...	...	...	...	...	...	...
248	Til Pitha	Glutinous rice, black sesame seeds, gur	vegetarian	30	sweet	dessert	Gujarat	North East
249	Bebinca	Coconut milk, egg yolks, clarified butter, all...	vegetarian	60	sweet	dessert	Gujarat	West
250	Shufta	Cottage cheese, dry dates, dried rose petals, ...	vegetarian	-1	sweet	dessert	Gujarat	North
251	Mawa Bati	Milk powder, dry fruits, arrowroot powder, all...	vegetarian	45	sweet	dessert	Gujarat	Central
252	Pinaca	Brown rice, fennel seeds, grated coconut, blac...	vegetarian	-1	sweet	dessert	Gujarat	West

253 rows × 8 columns

In [14]:

```
mode_state = df['state'].mode()[0]
df.loc[df['state'] == mode_state, 'region'] = mode_state + ' Region'
```

In [15]:

```
df
```

Out[15]:

	Name	Ingredients	Diet	cook_time	flavor_profile	course	state	region
0	Balu shahi	Maida flour, yogurt, oil, sugar	vegetarian	25	sweet	dessert	Gujarat	Gujarat Region
1	Boondi	Gram flour, ghee, sugar	vegetarian	30	sweet	dessert	Gujarat	Gujarat Region
2	Gajar ka halwa	Carrots, milk, sugar, ghee, cashews, raisins	vegetarian	60	sweet	dessert	Gujarat	Gujarat Region
3	Ghevar	Flour, ghee, kewra, milk, clarified butter, su...	vegetarian	30	sweet	dessert	Gujarat	Gujarat Region
4	Gulab jamun	Milk powder, plain flour, baking powder, ghee,...	vegetarian	40	sweet	dessert	Gujarat	Gujarat Region
...	...	...	...	...	...	...	...	...
248	Til Pitha	Glutinous rice, black sesame seeds, gur	vegetarian	30	sweet	dessert	Gujarat	Gujarat Region
249	Bebinca	Coconut milk, egg yolks, clarified butter, all...	vegetarian	60	sweet	dessert	Gujarat	Gujarat Region
250	Shufta	Cottage cheese, dry dates, dried rose petals, ...	vegetarian	-1	sweet	dessert	Gujarat	Gujarat Region
251	Mawa Bati	Milk powder, dry fruits, arrowroot powder, all...	vegetarian	45	sweet	dessert	Gujarat	Gujarat Region
252	Pinaca	Brown rice, fennel seeds, grated coconut, blac...	vegetarian	-1	sweet	dessert	Gujarat	Gujarat Region

253 rows × 8 columns

```
In [19]: df['cook_time'] = pd.to_numeric(df['cook_time'], errors='coerce')
mean_cook_time = df['cook_time'].mean()
df['cook_time'].fillna(mean_cook_time, inplace=True)
```

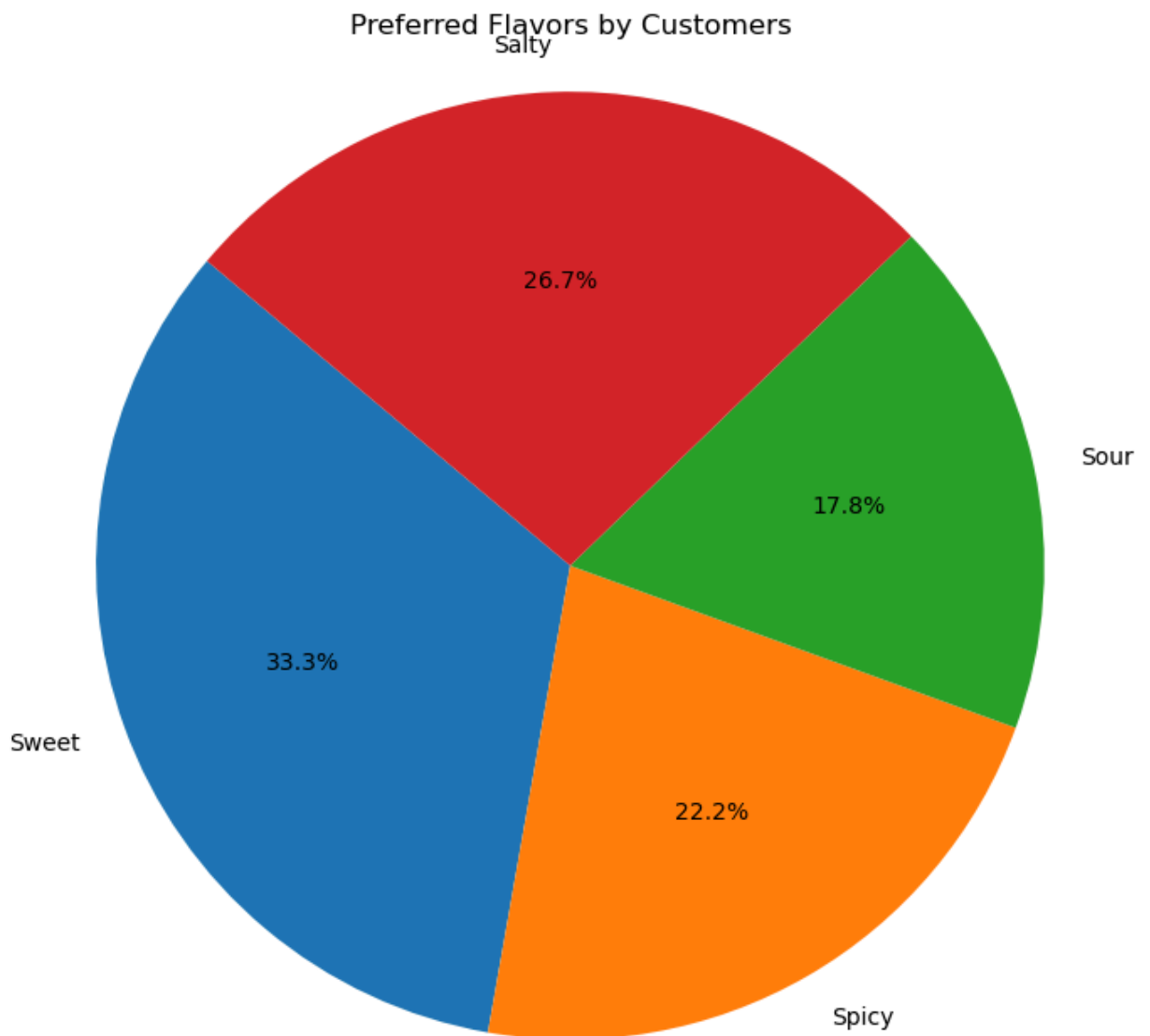
```
In [20]: df
```

Out[20]:

	Name	Ingredients	Diet	cook_time	flavor_profile	course	state	region
0	Balu shahi	Maida flour, yogurt, oil, sugar	vegetarian	25	sweet	dessert	Gujarat	Gujarat Region
1	Boondi	Gram flour, ghee, sugar	vegetarian	30	sweet	dessert	Gujarat	Gujarat Region
2	Gajar ka halwa	Carrots, milk, sugar, ghee, cashews, raisins	vegetarian	60	sweet	dessert	Gujarat	Gujarat Region
3	Ghevar	Flour, ghee, kewra, milk, clarified butter, su...	vegetarian	30	sweet	dessert	Gujarat	Gujarat Region
4	Gulab jamun	Milk powder, plain flour, baking powder, ghee,...	vegetarian	40	sweet	dessert	Gujarat	Gujarat Region
...	...	...	...	...	...	...	...	...
248	Til Pitha	Glutinous rice, black sesame seeds, gur	vegetarian	30	sweet	dessert	Gujarat	Gujarat Region
249	Bebinca	Coconut milk, egg yolks, clarified butter, all...	vegetarian	60	sweet	dessert	Gujarat	Gujarat Region
250	Shufta	Cottage cheese, dry dates, dried rose petals, ...	vegetarian	-1	sweet	dessert	Gujarat	Gujarat Region
251	Mawa Bati	Milk powder, dry fruits, arrowroot powder, all...	vegetarian	45	sweet	dessert	Gujarat	Gujarat Region
252	Pinaca	Brown rice, fennel seeds, grated coconut, blac...	vegetarian	-1	sweet	dessert	Gujarat	Gujarat Region

253 rows × 8 columns

```
In [25]: data = {
    'Flavor': ['Sweet', 'Spicy', 'Sour', 'Salty'],
    'Customers': [150, 100, 80, 120] # Example customer counts for each flavor
}
df_flavors = pd.DataFrame(data)
plt.figure(figsize=(8, 8))
plt.pie(df_flavors['Customers'], labels=df_flavors['Flavor'], autopct='%1.1f%%', start
plt.title('Preferred Flavors by Customers')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.show()
```



```
In [28]: data = {
          'Course': ['Main Course', 'Dessert', 'Appetizer', 'Soup'],
          'Customers': [200, 150, 120, 80]
        }

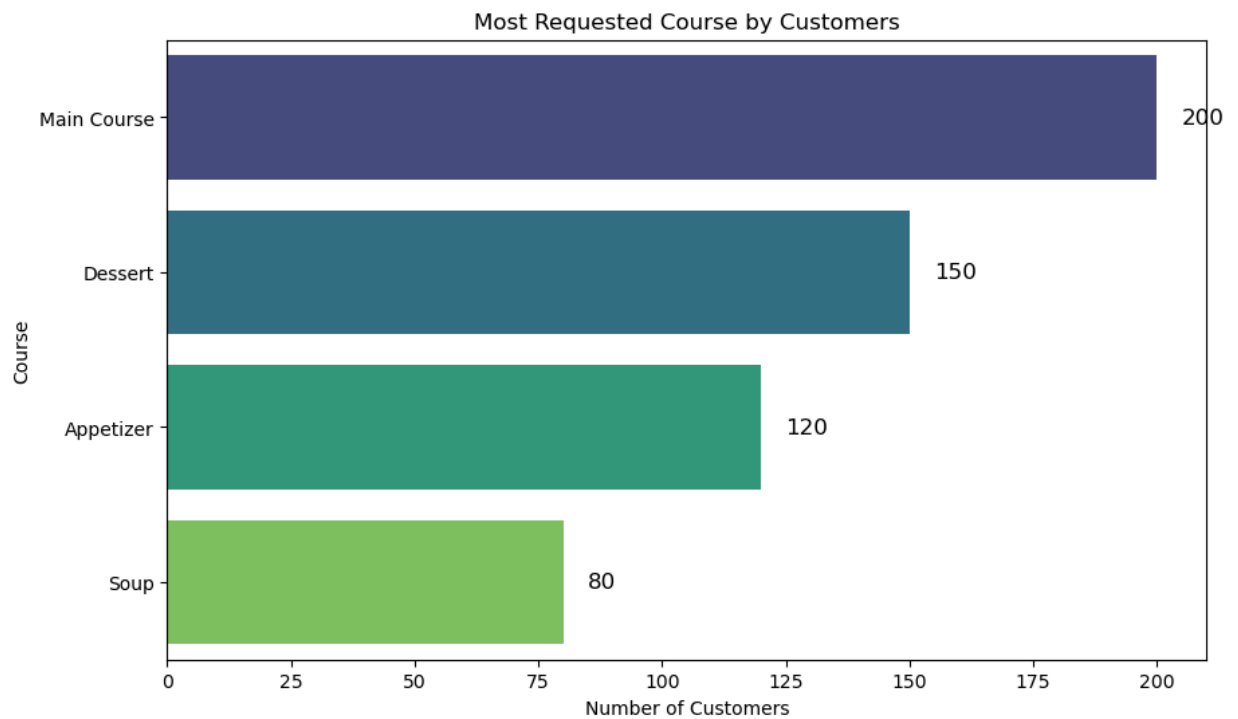
df_courses = pd.DataFrame(data)

df_courses = df_courses.sort_values(by='Customers', ascending=False)

plt.figure(figsize=(10, 6))
sns.barplot(x='Customers', y='Course', data=df_courses, palette='viridis')

for index, row in df_courses.iterrows():
    plt.text(row['Customers'] + 5, index, str(row['Customers']), va='center', fontsize=12)

plt.title('Most Requested Course by Customers')
plt.xlabel('Number of Customers')
plt.ylabel('Course')
plt.show()
```



QUESTION 2

In []:

In [39]: `import pandas as pd`

```
# Example file path, adjust according to your actual file location
file_path = 'Stats.csv'

# Attempt to read the CSV file with specified encoding
try:
    new_data = pd.read_csv(file_path, encoding='latin-1')
    print("CSV file loaded successfully.")
    # Proceed with data processing or analysis
except FileNotFoundError:
    print(f"Error: File '{file_path}' not found.")
except Exception as e:
    print(f"Error: {e}")
```

CSV file loaded successfully.

In [40]: `new_data`

Out[40]:

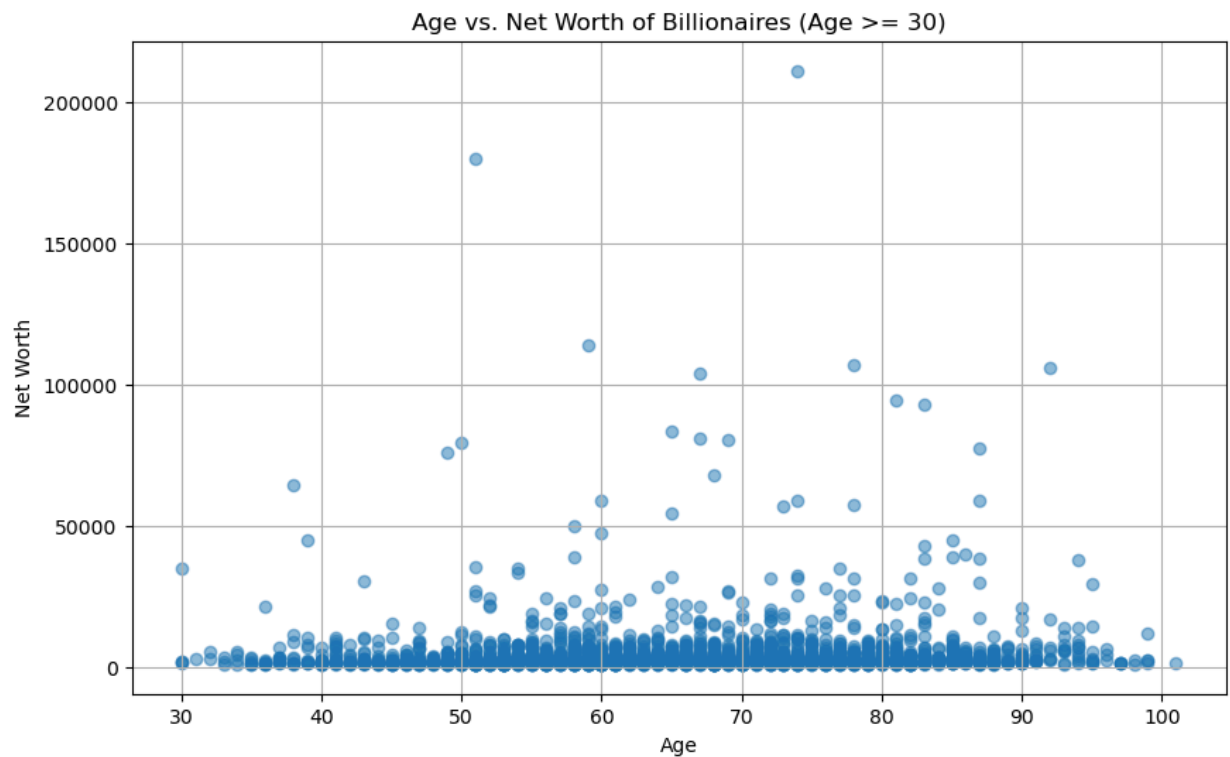
	finalWorth	category	personName	age	country	city	gender	birthYear	birthMo
0	211000	Fashion & Retail	Bernard Arnault & family	74.0	France	Paris	M	1949.0	
1	180000	Automotive	Elon Musk	51.0	United States	Austin	M	1971.0	
2	114000	Technology	Jeff Bezos	59.0	United States	Medina	M	1964.0	
3	107000	Technology	Larry Ellison	78.0	United States	Lanai	M	1944.0	
4	106000	Finance & Investments	Warren Buffett	92.0	United States	Omaha	M	1930.0	
...	...	...	...	...	...	...	...	...	
2635	1000	Healthcare	Yu Rong	51.0	China	Shanghai	M	1971.0	
2636	1000	Food & Beverage	Richard Yuengling, Jr.	80.0	United States	Pottsville	M	1943.0	
2637	1000	Manufacturing	Zhang Gongyun	60.0	China	Gaomi	M	1962.0	
2638	1000	Real Estate	Zhang Guiping & family	71.0	China	Nanjing	M	1951.0	
2639	1000	Diversified	Inigo Zobel	66.0	Philippines	Makati	M	1956.0	

2640 rows × 10 columns

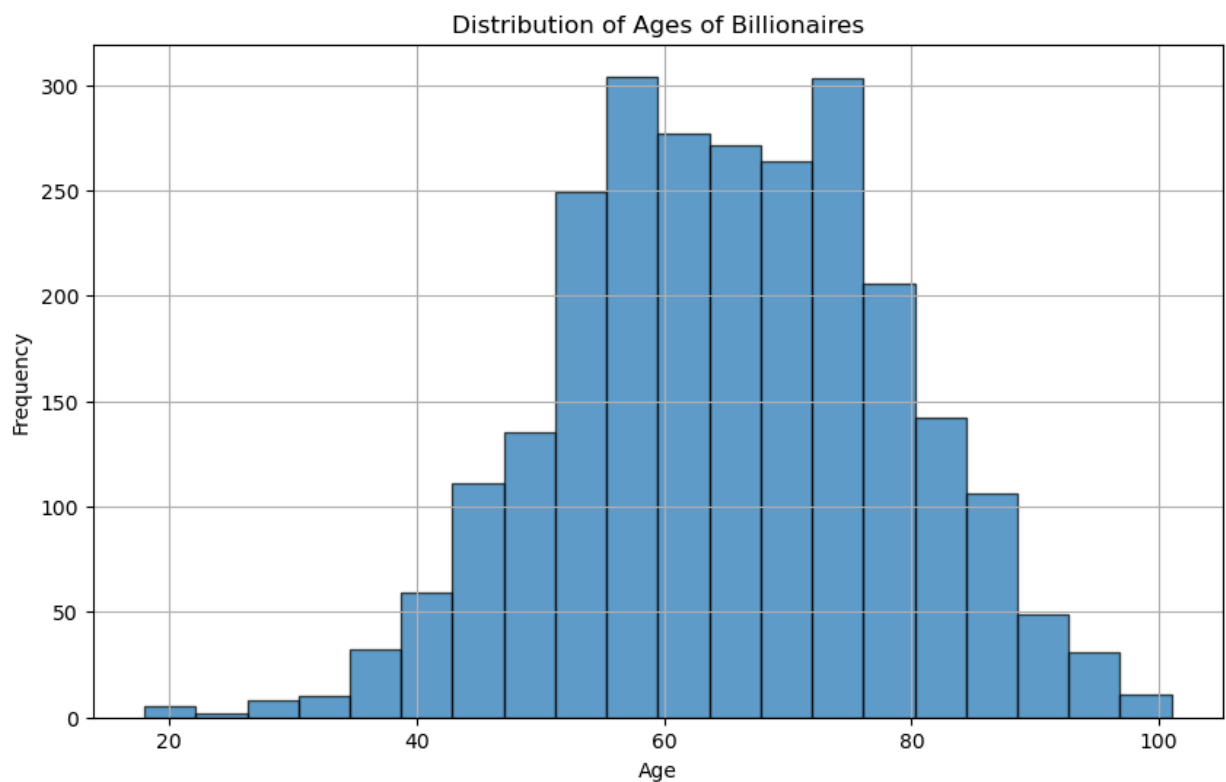
In [41]: `new_data.columns`

Out[41]: Index(['finalWorth', 'category', 'personName', 'age', 'country', 'city', 'gender', 'birthYear', 'birthMonth', 'birthDay'], dtype='object')

In [43]: `billionaires_over_30 = new_data[new_data['age'] >= 30]
plt.figure(figsize=(10, 6))
plt.scatter(billionaires_over_30['age'], billionaires_over_30['finalWorth'], alpha=0.5)
plt.title('Age vs. Net Worth of Billionaires (Age >= 30)')
plt.xlabel('Age')
plt.ylabel('Net Worth')
plt.grid(True)
plt.show()`



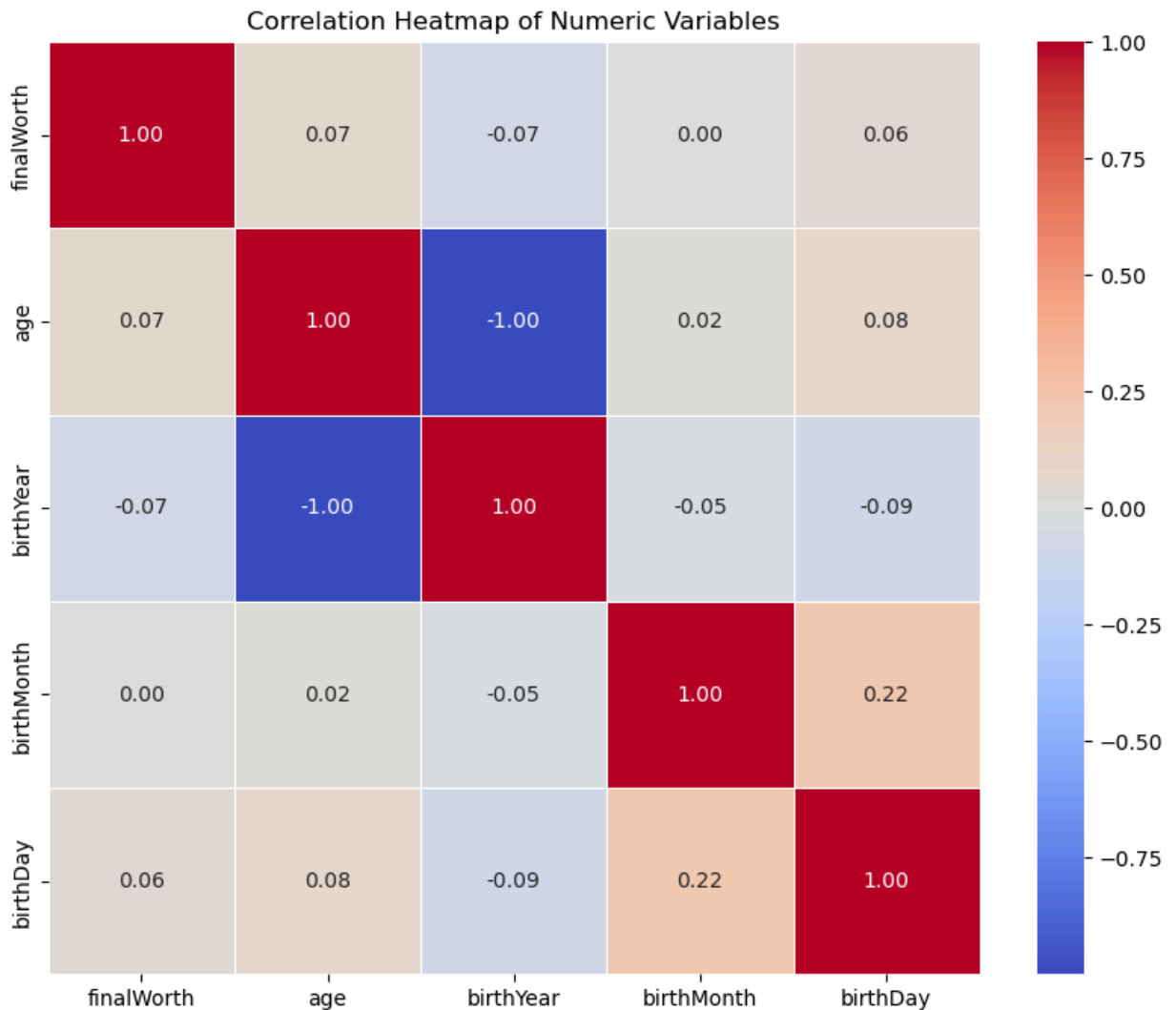
```
In [44]: plt.figure(figsize=(10, 6))
plt.hist(new_data['age'], bins=20, edgecolor='black', alpha=0.7)
plt.title('Distribution of Ages of Billionaires')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



```
In [45]: numeric_columns = new_data.select_dtypes(include=['float64', 'int64'])
```

```
correlation_matrix = numeric_columns.corr()

plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', square=True, fmt='.2f',
            plt.title('Correlation Heatmap of Numeric Variables')
plt.show()
```



```
In [46]: plt.savefig('correlation_heatmap.png')
```

<Figure size 640x480 with 0 Axes>

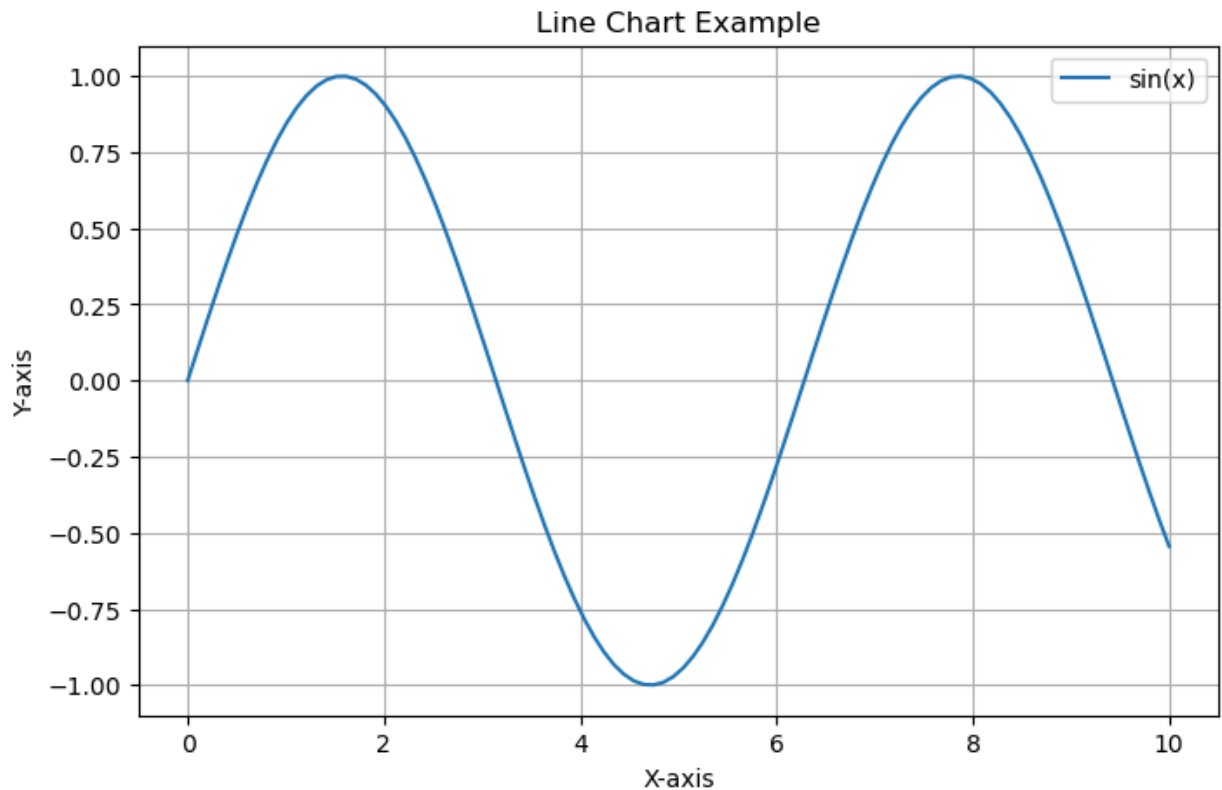
QUESTION 3

Line Chart A line chart is used to display data points connected by straight line segments.

```
In [47]: import matplotlib.pyplot as plt
import numpy as np
```

```
x = np.linspace(0, 10, 100)
y = np.sin(x)
```

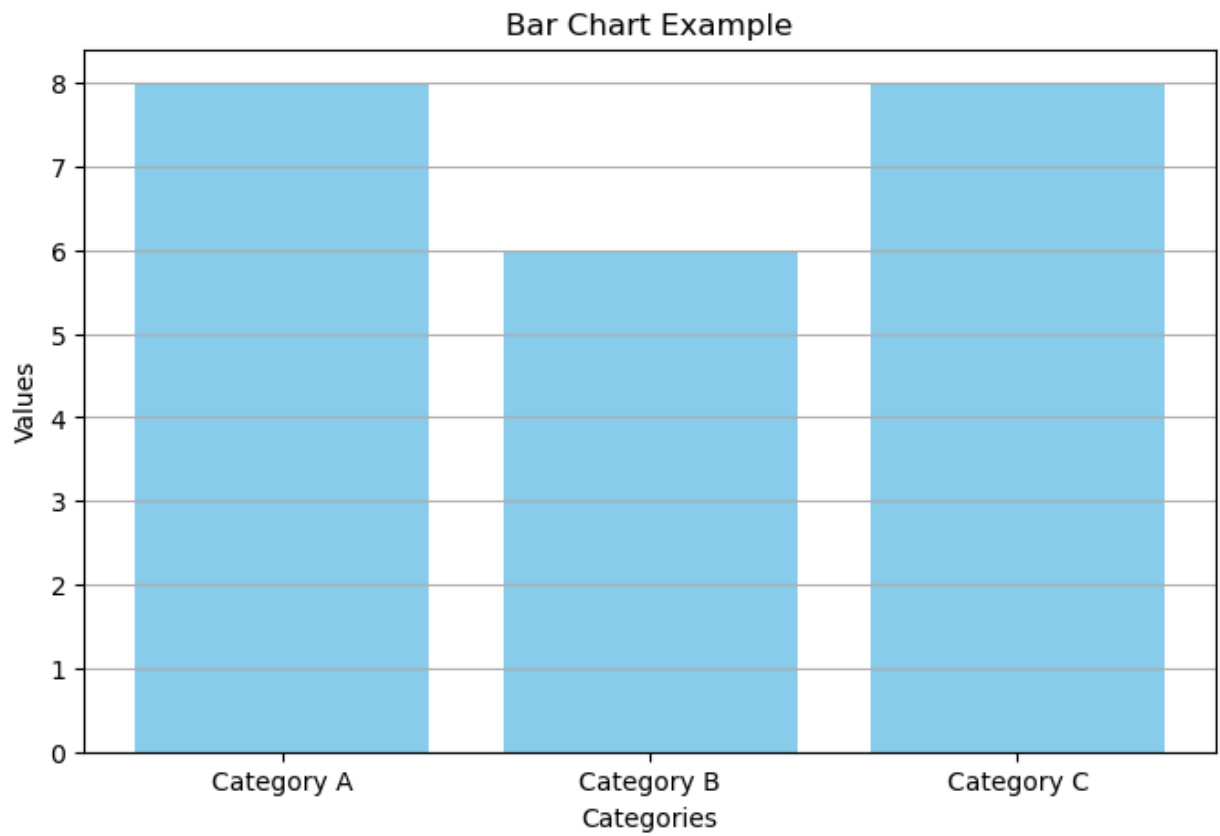
```
plt.figure(figsize=(8, 5))
plt.plot(x, y, label='sin(x)')
plt.title('Line Chart Example')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.legend()
plt.grid(True)
plt.show()
```



Bar Chart A bar chart represents categorical data with rectangular bars, where the length of each bar is proportional to the value it represents.

```
In [48]: categories = ['Category A', 'Category B', 'Category C']
values = np.random.randint(1, 10, size=len(categories))

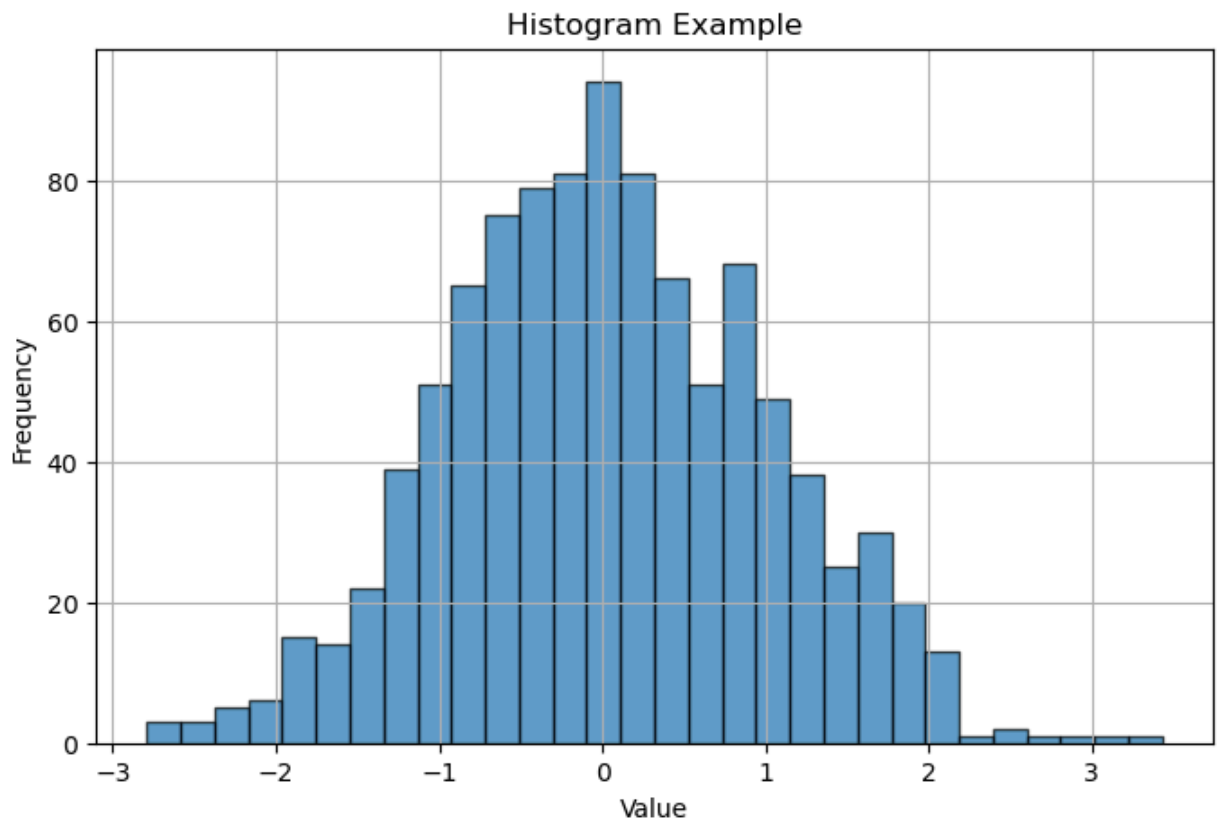
plt.figure(figsize=(8, 5))
plt.bar(categories, values, color='skyblue')
plt.title('Bar Chart Example')
plt.xlabel('Categories')
plt.ylabel('Values')
plt.grid(axis='y')
plt.show()
```



Histogram A histogram displays the distribution of a dataset by dividing it into bins and counting the number of observations in each bin.

```
In [49]: data = np.random.normal(loc=0, scale=1, size=1000)

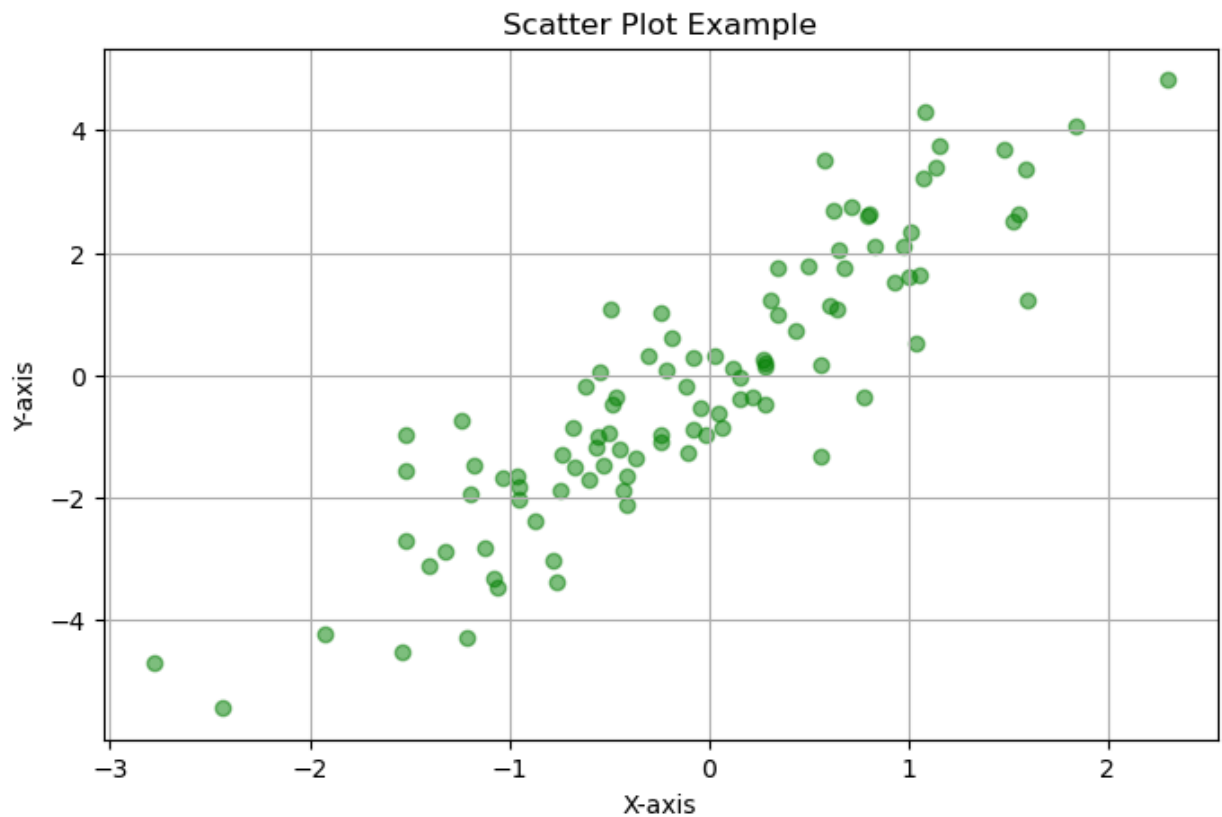
plt.figure(figsize=(8, 5))
plt.hist(data, bins=30, edgecolor='black', alpha=0.7)
plt.title('Histogram Example')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



Scatter Plot A scatter plot shows the relationship between two numerical variables by displaying data points as dots.

```
In [50]: x = np.random.randn(100)
y = 2 * x + np.random.randn(100)

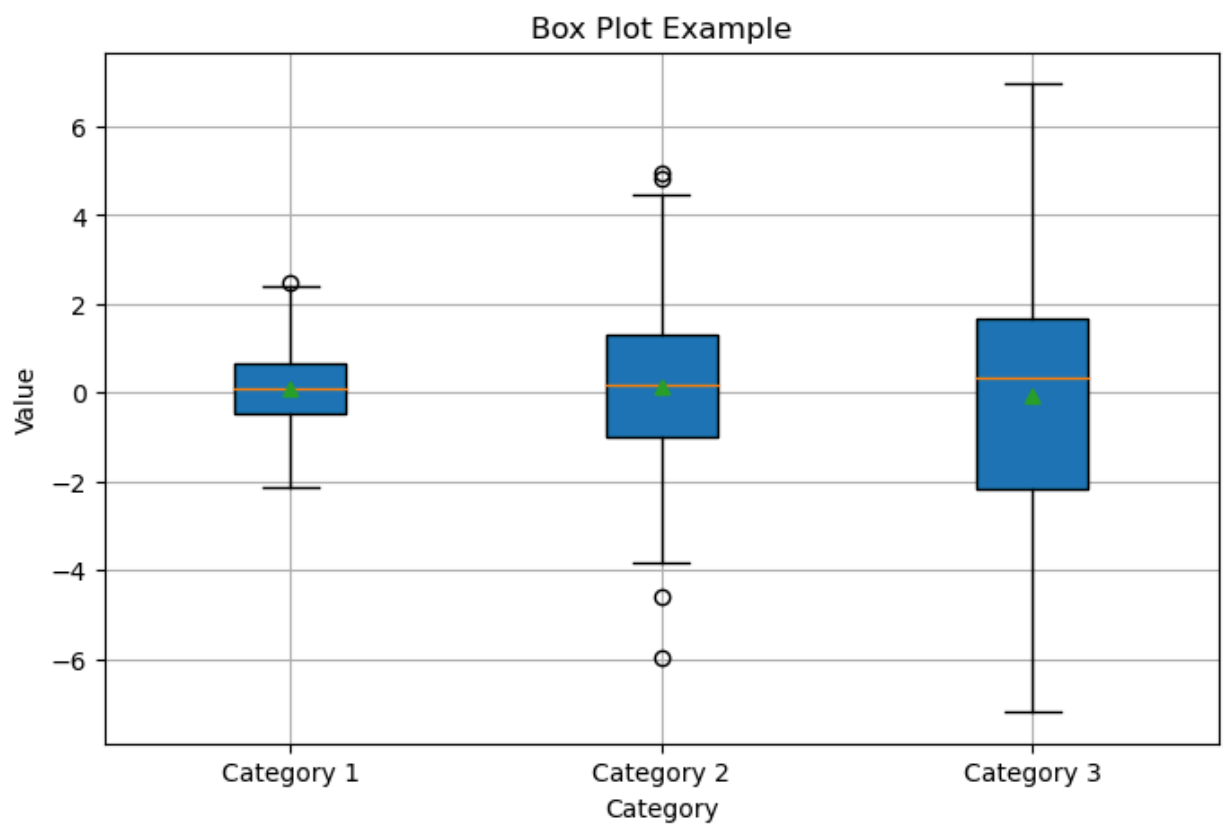
plt.figure(figsize=(8, 5))
plt.scatter(x, y, color='green', alpha=0.5)
plt.title('Scatter Plot Example')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.grid(True)
plt.show()
```



Box Plot A box plot (or box-and-whisker plot) provides a summary of the distribution of a dataset, showing median, quartiles, and outliers.

```
In [51]: np.random.seed(10)
data = [np.random.normal(0, std, 100) for std in range(1, 4)]

plt.figure(figsize=(8, 5))
plt.boxplot(data, patch_artist=True, showmeans=True)
plt.title('Box Plot Example')
plt.xlabel('Category')
plt.ylabel('Value')
plt.xticks([1, 2, 3], ['Category 1', 'Category 2', 'Category 3'])
plt.grid(True)
plt.show()
```



In []: